

PROJECTIVE METABELIAN GROUPS AND LIE ALGEBRAS

UDC 519.4

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Abstract. Suppose that A_n is the variety of all abelian groups of exponent dividing $n \geq 0$, and $A_0 = A$ is the variety of all abelian groups. In this paper it is proved that projective metabelian $A_n A_m$ -groups of finite rank are free. Moreover, it is proved that projective metabelian $k[Y_1^{\pm 1}, \dots, Y_r^{\pm 1}, Z_1, \dots, Z_s]$ -Lie algebras of finite rank, where k is a principal ideal ring, are free.

Bibliography: 9 titles.

Let A_n be the variety of all abelian groups of exponent dividing $n \geq 0$, and let $A_0 = A$ be the variety of all abelian groups. The present paper is devoted to a problem of P. Hall (see [1], Problem 21), as to whether projective metabelian $A_n A_m$ -groups of finite rank are free. If n and m are greater than 0 and nm is not primary, then a one-generator free $A_n A_m$ -group is isomorphic to $Z/(nm)$ and therefore has a nontrivial direct decomposition; that is, there are nonfree projective groups in this case. But if $nm = p^a$, where p is prime, then $A_n A_m$ is locally nilpotent, and therefore projective $A_n A_m$ -groups S of finite rank

$$r = \dim_{F_p} S / \{S', S^p\}$$

are free in this case ([1], Corollary 44.33).

Thus only the cases $n = 0$ or $m = 0$ are nontrivial. In both cases $A_n A_m \supset A$, and therefore, if P is projective in $A_n A_m$, P/P' is a free abelian group. We define the rank $r = r(P) = \text{rank } P/P'$. It is shown in [2] that if $n = 0$ and $m = r = 2$, then P is free in AA_2 . But if $r, m \geq 2$ and $n = 0$, except for $m = r = 2$, then there are nonfree projective $A_n A_m$ -groups of rank r , examples of which are constructed in [3].

In the present paper we show that projective $A_n A$ -groups for $n \geq 0$, and projective metabelian $k[Y_1^{\pm 1}, \dots, Y_r^{\pm 1}, Z_1, \dots, Z_s]$ -Lie algebras of finite rank, where k is a principal ideal ring (see [4] and [5]), are free. It follows that all projective restricted Lie algebras of finite rank over a field k are free. The proof of these results depends on the positive solution of Serre's problem obtained by Suslin, Quillen and Swan.

§1. Free $A_n A$ -groups

Suppose that G is a free metabelian A^2 -group with the basis z_i , $i \in I$. We write Z_i for the coset of z_i in $B = G/G'$. If T_0 is a free ZB -module with the basis w_i , $i \in I$, then we define an

AMS (MOS) subject classifications (1970). Primary 20E10, 20E15; Secondary 20E05, 17B30.

epimorphism of $\mathbf{Z}B$ -modules $l_0: T_0 \rightarrow m_0$, where m_0 is the augmentation ideal of $\mathbf{Z}B$. We set $l_0(w_i) = Z_i - 1 \in m_0$. As is shown in [1], Chapter III, §6, and in [8], G is isomorphic to the group of all matrices

$$\begin{pmatrix} p & 0 \\ d & 1 \end{pmatrix}, \quad p \in B, \quad d \in T_0, \quad (1)$$

with $l_0(d) = p - 1$, and the matrices

$$z_i = \begin{pmatrix} Z_i & 0 \\ w_i & 1 \end{pmatrix} \quad (1')$$

are free generators of G . We shall therefore henceforth take G to be the group of matrices of the form (1) with the free generators (1').

Suppose that $n \geq 0$. It is clear that the verbal subgroup $V \triangleleft G$ corresponding to the subvariety $A_n A$ consists of the matrices of the form

$$V = \left\{ \begin{pmatrix} 1 & 0 \\ nd & 1 \end{pmatrix} \mid d \in \text{Ker } l_0 \subset T_0 \right\}. \quad (1'')$$

THEOREM 1. *Let B be a free abelian group with the basis Z_i , $i \in I$, and let T be a free $\mathbf{Z}/(n)B$ -module with the basis w_i , $i \in I$, where $\mathbf{Z}/(n)B$ is the group algebra of B with coefficients in $\mathbf{Z}/(n)$. If m is the augmentation ideal of $\mathbf{Z}/(n)B$, and $l: T \rightarrow m$ is the epimorphism of $\mathbf{Z}/(n)B$ -modules given by $l(w_i) = Z_i - 1$, then the group F of all matrices*

$$\begin{pmatrix} p & 0 \\ d & 1 \end{pmatrix},$$

where $p \in B$, $d \in T$ and $l(d) = p - 1$, is a free $A_n A$ -group with the basis

$$z_i = \begin{pmatrix} Z_i & 0 \\ w_i & 1 \end{pmatrix}. \quad (1''')$$

PROOF. By (1)–(1''), the free $A_n A$ -group G/V is contained in F and is freely generated by the elements (1'''). Hence it remains to show that F is in fact generated by the elements z_i of (1'''). Let

$$a = \begin{pmatrix} p & 0 \\ d & 1 \end{pmatrix} \in F.$$

Because p is a product of the $Z_i^{\pm 1}$, we can, by multiplying a by a product of the $z_i^{\pm 1} \in G/V$, assume that

$$a = \begin{pmatrix} 1 & 0 \\ d & 1 \end{pmatrix} \in F,$$

where $l(d) = 0$. Suppose that $d = \sum b_i w_i$, where $b_i \in \mathbf{Z}/(n)B$. Then, by hypothesis, $l(d) = \sum b_i (Z_i - 1) = 0$. Lifting to T_0 and $\mathbf{Z}B$, we obtain

$$l_0(d) = \sum b_i (Z_i - 1) \in n \mathbf{Z} B, \quad \sum b_i (Z_i - 1) = nf, \quad f \in m_0,$$

for $d = \sum b_i w_i \in T_0$; that is, $f = \sum c_i (Z_i - 1)$ and $\sum (b_i - nc_i)(Z_i - 1) = 0$, whence

$$d = \begin{pmatrix} 1 & 0 \\ \sum (b_i - nc_i) w_i & 1 \end{pmatrix} \in G,$$

and therefore $a \in G/V$. This proves the theorem.

THEOREM 2. *Suppose that all projective $A_q A$ -groups of rank r are free for all primes q . Then projective $A_n A$ -groups of rank r are free for every $n > 0$.*

PROOF. Suppose that P is a projective $A_n A$ -group of rank r . Then the derived group P' is a $\mathbb{Z}/(n)B_r$ -module, where $B_r = P/P'$, and if q is prime and q^k divides n , then $q^k P'$ is the verbal subgroup of P corresponding to the subvariety $A_{q^k} A \subseteq A_n A$. This follows from (1'') owing to the fact that if P is a retract of an $A_n A$ -free group F , then we have $V(P) = V(F) \wedge P$ for the verbal subgroup $V(F)$. By hypothesis P/qP' is a free $A_q A$ -group of rank r . Because every automorphism of P/P' is induced by an automorphism of P/qP' (see [1], Theorem 41.21), there are elements $a_1(q), \dots, a_r(q)$ of P for every $q|n$, such that $a_i(q)$ is independent of q modulo P' and $a_1(q), \dots, a_r(q)$ are free generators of P/qP' .

LEMMA. *There are $a_1, \dots, a_r \in P$ such that*

$$a_i \equiv a_i(q_j) \pmod{q_j^{k_j} P'}, \quad n = \prod_j q_j^{k_j}.$$

PROOF. Suppose that P is a retract of the free $A_n A$ -group F . We consider the $\mathbb{Z}/(n)B$ -module T associated with F by Theorem 1, where $B = F/F'$. If $n = \prod_j q_j^{k_j}$, then $1 = \sum e_j$, where $e_j \in \mathbb{Z}$ and

$$e_j \equiv 1 \pmod{q_j^{k_j}}, \quad e_j \in (nq_j^{-k_j}).$$

Then $T_j = e_j T$ is the $\mathbb{Z}/(q_j^{k_j})B$ -module associated with the free $A_{q_j^{k_j}} A$ -group F_j on the same set of free generators as F . Here $T = \bigoplus_j T_j$ and the natural projection of T onto T_j is multiplication by e_j . Moreover, we have $l|_{T_j} = e_j l_j$ for the epimorphisms l and l_j from T and T_j onto the augmentation ideals $\mathbb{Z}/(n)B$ and $\mathbb{Z}/(q_j^{k_j})B$. Suppose that

$$l(q_j) = \begin{pmatrix} c_i & 0 \\ d_{ij} & 1 \end{pmatrix} \in P/q_j^{k_j} P', \quad 1 \leq i \leq r,$$

where $d_{ij} \in T_j = e_j T$ and $l(d_{ij}) = e_j(c_i - 1)$. We set

$$b_i = \begin{pmatrix} c_i & 0 \\ \sum_j d_{ij} & 1 \end{pmatrix}.$$

Then $b_i \in F$, because

$$l\left(\sum_j d_{ij}\right) = \sum_j l(d_{ij}) = \sum_j e_j l_j(d_{ij}) = \left(\sum_j e_j\right)(c_i - 1) = c_i - 1.$$

We set $a_i = b_i^\pi \in P$. To prove the lemma it is enough to show that $a_i \equiv a_i(q_j) \pmod{q_j^{k_j} P'}$. By construction, $a_i(q_j) \equiv b_i \pmod{q_j^{k_j} P'}$, whence

$$a_i(q_j) \equiv a_i(q_j)^\pi \equiv b_i^\pi = a_i \bmod q_j^{k_j} P'.$$

This proves the lemma.

To complete the proof of the theorem, it is enough to show that the commutators $[a_i, a_j]$, $1 \leq i, j \leq r$, generate P' as a $\mathbf{Z}/(n)B$ -module. Then $a_1, \dots, a_r$ generate P and it is therefore free in $A_n A$.

So we have $P' = \bigoplus_j e_j P'$, where the e_j are as in the lemma. Here

$$q_i P' = \bigoplus_{j \neq i} e_j P' \oplus q_i e_i P', \quad P'/q_i P' = e_i P'/q_i e_i P'.$$

Because $e_j[a_i, a_k] = [a_i(q_j), a_k(q_j)]$, the submodule generated by the $[a_i, a_k]$, $1 \leq i, k \leq r$, contains $e_j P'$ by Nakayama's lemma and therefore also contains P' . This proves the theorem.

Thus we shall henceforth study projective $A_q A$ -groups of finite rank, where q is either prime or zero. Suppose that F is a free $A_q A$ -group, and P is a retract of F with a projection $\pi: F \rightarrow P$, while $r(P) = r$. We choose a basis of F compatible with π . To do this, we remark that P/P' is a direct factor of the free abelian group F/F' . Therefore there is a basis $x_1, \dots, x_r, y_j, j \in I$, of F such that

$$x_i^\pi \equiv x_i \bmod F', \quad y_j^\pi \in F', \quad j \in I.$$

The number r is, as is not hard to see, the rank of P , since r is the rank of P/P' . Henceforth we shall assume that $r < \infty$.

Suppose, as above, that $B = F/F'$ and $B_r = P/P' \subseteq B$. We denote the images of x_i and y_j in B by X_i and Y_j . Then B is a free abelian group with the basis X_i, Y_j , and B_r is a free abelian group with the basis $X_1, \dots, X_r$. Suppose that T is a free $\mathbf{Z}/(q)B$ -module with the basis $u_1, \dots, u_r, v_j, j \in I$, and M is a free $\mathbf{Z}/(q)B_r$ -module with the same basis. Then

$$T = \mathbf{Z}/(q) B \bigotimes_{\mathbf{Z}/(q)B_r} M$$

and therefore $T \supseteq M$. We define a map $\zeta: T \rightarrow M$ as follows: for

$$f = \sum_{i=1}^r f_i(X, Y) u_i + \sum_{j \in I} f_j(X, Y) v_j, \quad f_i, f_j \in \mathbf{Z}/(q) B,$$

we set

$$f^\zeta = \sum_{i=1}^r f_i(X, 1) u_i + \sum_{j \in I} f_j(X, 1) v_j.$$

It is clear that ζ is an epimorphism of $\mathbf{Z}/(q)B_r$ -modules, and $\zeta|_M$ is the identity. By Theorem 1 (see also [1] and [8]), F can be considered as the group of affine transformations of T generated by

$$x_i = \begin{pmatrix} X_i & 0 \\ u_i & 1 \end{pmatrix}, \quad y_j = \begin{pmatrix} Y_j & 0 \\ v_j & 1 \end{pmatrix}. \quad (2)$$

Here, if $l': T \rightarrow m'$ is the epimorphism of the $\mathbf{Z}/(q)B$ -module T onto the augmentation ideal $m' \triangleleft \mathbf{Z}/(q)B$ given by $l'(u_i) = X_i - 1$ and $l'(v_j) = Y_j - 1$, then, for $a \in B$ and $b \in T$,

$$z = \begin{pmatrix} a & 0 \\ b & 1 \end{pmatrix} \in F \quad (2')$$

if and only if

$$l'(b) = a - 1. \quad (2'')$$

Here z in (2') lies in F' if and only if $a = 1$; that is, $l(b) = 0$. We remark that if

$$t_i = \begin{pmatrix} a_i & 0 \\ b_i & 1 \end{pmatrix} \in F, \quad i = 1, 2, \quad (3)$$

then, as in [3], p. 111,

$$t_1 t_2 = \begin{pmatrix} a_1 a_2 & 0 \\ a_2 b_1 + b_2 & 1 \end{pmatrix}, \quad t_1^{-1} = \begin{pmatrix} a_1^{-1} & 0 \\ -a_1^{-1} b_1 & 1 \end{pmatrix}, \quad (3')$$

$$t_1' = t_2^{-1} t_1 t_2 = \begin{pmatrix} a_1 & 0 \\ a_2 b_1 - (a_1 - 1) b_2 & 1 \end{pmatrix}. \quad (3'')$$

Therefore Lemmas 1 and 2 on p. 113 of [3] hold for F . In particular, we have

PROPOSITION 1. *Let*

$$U = \left\{ \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \in F \mid c \in \text{Ker } \xi \right\}.$$

Then $U \subseteq \text{Ker } \pi$.

Henceforth, for the sake of simplicity, we shall write E_r for $\mathbf{Z}/(q)B_r$. The group $H = F/U$ acts as a group of affine transformations of the E_r -module M , and multiplication by Y_i is the identity on M . Therefore H is generated by the matrices

$$x_i = \begin{pmatrix} X_i & 0 \\ u_i & 1 \end{pmatrix}, \quad y_j = \begin{pmatrix} 1 & 0 \\ v_j & 1 \end{pmatrix}$$

and the multiplication in the subgroup generated by these matrices is given by (3)–(3''). We define an epimorphism $l: M \rightarrow m_r$, where m_r is the augmentation ideal of E_r , by setting $l(u_i) = X_i - 1$ and $l(v_j) = 0$. We remark that if ξ_1 is the projection of B onto B_r , then

$$l(b^{\xi_1}) = a^{\xi_1} - 1 \quad (4)$$

for $\begin{pmatrix} a & 0 \\ b & 1 \end{pmatrix} \in F$.

PROPOSITION 2. *The matrix*

$$z = \begin{pmatrix} f & 0 \\ g & 1 \end{pmatrix}, \quad f \in B_r, \quad g \in M,$$

belongs to H *if and only if* $l(g) = f - 1$. *In particular, z belongs to the subgroup generated by* H' *and the* y_j *if and only if* $l(g) = 0$.

PROOF. By (4), $z \in H$ implies $l(g) = f - 1$. Conversely, suppose we are given

$$z = \begin{pmatrix} f & 0 \\ g & 1 \end{pmatrix}, \quad f \in B_r, \quad g \in M,$$

where $f - 1 = l(g)$. Multiplying by a product of the x_i , we may suppose that $f = 1$. Then

$$g = \sum g_i u_i + \sum g'_j v_j, \quad g_i, g'_j \in E_r,$$

$$l(g) = \sum g_i (X_i - 1) = 0.$$

Hence, by the criterion for F , we have

$$\begin{pmatrix} 1 & 0 \\ \sum g_i u_i & 1 \end{pmatrix} \in H.$$

It follows from (3)–(3'') that

$$\begin{pmatrix} 1 & 0 \\ \sum g'_j v_j & 1 \end{pmatrix} = \prod_j \begin{pmatrix} 1 & 0 \\ g'_j v_j & 1 \end{pmatrix} \in H.$$

This proves the proposition.

Because $U \subseteq \text{Ker } \pi$, π induces an idempotent map π in H and P is a retract of H with the projection π . We define an endomorphism ρ of M by setting

$$x_i^\pi = \begin{pmatrix} X_i & 0 \\ u_i^\rho & 1 \end{pmatrix}, \quad y_j^\pi = \begin{pmatrix} 1 & 0 \\ v_j^\rho & 1 \end{pmatrix}.$$

We remark that this is well-defined, since $x_i^\pi \equiv x_i \pmod{H'}$ and $y_j^\pi \in H'$.

PROPOSITION 3. *If*

$$t = \begin{pmatrix} a & 0 \\ b & 1 \end{pmatrix} \in H,$$

then

$$t^\pi = \begin{pmatrix} a & 0 \\ b^\rho & 1 \end{pmatrix},$$

whence $\rho^2 = \rho$.

PROOF. Suppose that the assertion holds for

$$t_i = \begin{pmatrix} a_i & 0 \\ b_i & 1 \end{pmatrix}, \quad i = 1, 2,$$

Then by (3)–(3'') it also holds for t_{12} and t^{-1} . Hence $u_i^{\rho^2} = u_i^\rho$, $v_j^{\rho^2} = v_j^\rho$ and $\rho^2 = \rho$.

COROLLARY. *If $b \in M$, then $l(b) = l(b^\rho)$.*

Let $S = \text{Im } \rho$. By Proposition 3, S is a projective E_r -module with an epimorphism $l: S \rightarrow \mathfrak{m}_r$, since $l(u_i^\rho) = l(u_i) = X_i - 1$. We compute $\text{Ker } l$.

PROPOSITION 4. *$\text{Ker } l$ is generated as an E_r -module by the elements v_j and $u_i(X_j - 1) - u_j(X_i - 1)$.*

PROOF. We may assume that $c \in \text{Ker } l$ belongs to $E_r \mu_1 \oplus \dots \oplus E_r \mu_r$. Then

$$h = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \in H'$$

and therefore

$$h = \prod [x_i^{\varepsilon_i}, x_j^{\varepsilon_j}]^{B_{ij}}, \quad B_{ij} \in E_r, \quad \varepsilon_i, \varepsilon_j = \pm 1.$$

By (3)–(3''), we have

$$c = \sum [(p_j - 1) q_i - (p_i - 1) q_j] B_{ij},$$

where

$$x_i^{\varepsilon_i} = \begin{pmatrix} p_i & 0 \\ q_i & 1 \end{pmatrix}, \quad x_j^{\varepsilon_j} = \begin{pmatrix} p_j & 0 \\ q_j & 1 \end{pmatrix}.$$

But $q_i(p_j - 1) - (p_i - 1)q_j$ belongs to the submodule generated by the $u_i(X_j - 1) - u_j(X_i - 1)$. This proves the proposition.

§2. Auxiliary statements

In §1 we constructed a projective E_r -module S , with an epimorphism $l: S \rightarrow \mathfrak{m}$, from a projective group P of rank r . In this section we shall obtain some statements, which are generalizations of results in [4] and [5], concerning the existence of a free basis of S connected with l .

Generalizing the situation in §1, we consider a chain of commutative rings $E_0 \subset E_1 \subset \dots$ with the following properties:

- 1) The unit of E_r lies in E_0 for all r .
- 2) Each E_r is a retract of E_{r+1} with kernel (X_{r+1}) , where X_{r+1} is not a divisor of zero in $\bigcup E_i$.
- 3) $\text{GL}(t, E_r)$ acts transitively on the set of unimodular vectors in E_r for all $t \geq r$.
- 4) If $\mathfrak{m}_r = (X_1, \dots, X_r) \triangleleft E_r$, then $\mathfrak{m}_r/\mathfrak{m}_r^2$ is a free E_0 -module of rank r .

We remark that E_0 is a retract of E_r with kernel $\mathfrak{m}_r$; that is, $E_r = E_0 + \mathfrak{m}_r$, and therefore the elements $X_i + \mathfrak{m}_r^2$, $1 \leq i \leq r$, form a basis of $\mathfrak{m}_r/\mathfrak{m}_r^2$ in 4).

Suppose that E_r^t is a free E_r -module of rank $t \geq r$. If $f = [f_1, \dots, f_t] \in E_r^t$, $f_i \in E_r$, we set $(f) = (f_1, \dots, f_t)$. We introduce the symmetric bilinear function

$$\langle f, g \rangle = \sum_{i=1}^t f_i g_i$$

on E_r^t , where $f = [f_1, \dots, f_t]$ and $g = [g_1, \dots, g_t]$. We remark that

$$\langle f, g^\alpha \rangle = \langle f^\alpha, g \rangle \quad (5)$$

for $\alpha \in \text{GL}(t, E_r)$. We set $X(r) = [X_1, \dots, X_r, 0, \dots, 0]$ and, following [4] and [5], we call a vector $\mathfrak{m}_r$ -modular if $(f) = \mathfrak{m}_r$.

PROPOSITION 5. Suppose that $f \in E_r$ with coordinates

$$f_i = X_i + a_i X_r, \quad 1 \leq i \leq r-1,$$

$$f_r = a_0 X_r + a_r X_r^2, \quad f_j = X_r^2 a_j, \quad j > r,$$

is $\mathfrak{m}_r$ -modular, $a_0 \in E_0$ and $a_i \in E_i$, $i \geq 1$. Then a_0 is invertible in E_0 .

PROOF. Because $(f) = \mathfrak{m}_r$, there is a $b = [b_1, \dots, b_t] \in E_r^t$ such that $X_r = \langle f, b \rangle$. But $E_r = E_{r-1} + (X_r)$ and therefore

$$b_i = b'_i + X_r c_i, \quad b'_i \in E_{r-1}, \quad c_i \in E_r.$$

Modulo X_r^2 we have

$$X_r \equiv X_r a_0 b'_r + \sum_{i=1}^{r-1} X_i b'_i + X_r \sum_{i=1}^{r-1} (X_i c_i + a_i b'_i). \quad (6)$$

Because $E_{r-1} \wedge (X_r) = 0$, (6) implies $\sum_{i=1}^{r-1} X_i b'_i = 0$; that is, $b'_i \in \mathfrak{m}_r$, since $\mathfrak{m}_r/\mathfrak{m}_r^2$ is a free E_0 -module. Thus, we have modulo $\mathfrak{m}_r^2$ in (6)

$$X_r = X_r a_0 b''_r, \quad b''_r \equiv b_r \pmod{\mathfrak{m}_{r-1}}, \quad b''_r \in E_0.$$

By (4) we obtain that $a_0 b''_r = 1$. This proves the proposition.

PROPOSITION 6. Suppose given an $\mathfrak{m}_r$ -modular vector $f \in E_r^t$, $f \equiv X(r) \pmod{X_r}$. Then there is an $\alpha \in \mathbf{GL}(t, E_r)$ such that $f^\alpha \equiv X(r) \pmod{X_r^2}$. If $g \in E_1^t$ is $\mathfrak{m}_1$ -modular, then $g^\alpha \equiv X(1) \pmod{X_1^2}$ for some $\alpha \in \mathbf{GL}(t, E_1)$.

PROOF. Suppose that $f = [f_1, \dots, f_t] \equiv X(r) \pmod{X_r}$; that is, $f_i = X_i + X_r h_i$ for $1 \leq i \leq r$, and $f_j = X_r h_j$ for $j > r$. We suppose that $h_j \equiv c_j \pmod{\mathfrak{m}_r}$, $c_j \in E_0$. Because f is $\mathfrak{m}_r$ -modular, we have

$$X_r \equiv \left[\sum_{i=1}^r (X_i + X_r c_i) b_i + \sum_{i>r} X_r c_i b_i \right] \pmod{(\mathfrak{m}_r^2 \wedge (X_r))},$$

for some $b_1, \dots, b_t \in E_0$, whence, as in Proposition 5,

$$1 = b_r + \sum_{i \geq r \geq 1} c_i b_i, \quad \sum_{i=1}^{r-1} X_i b_i = 0;$$

that is, $[b_1, \dots, b_r] \in E_0^t$ is unimodular. We remark that $\mathfrak{m}_r^2 \wedge (X_r) = X_r \mathfrak{m}_r$ by (4). By 3), $b_1, \dots, b_t$ can be made to form the r th column of a matrix $\beta \in \mathbf{GL}(t, E_0)$. Applying β to the row made up of the coordinates of f , we obtain that the coordinates of $g = f^\beta$ are of the form

$$g_i \equiv \sum_{j=1}^r \beta_{ij} X_j \pmod{X_r \mathfrak{m}_r}, \quad i \neq r,$$

$$g_r \equiv X_r \pmod{X_r \mathfrak{m}_r}.$$

Applying elementary transformations in $\mathbf{GL}(r-1, E_0)$ to g , we may assume, because of 3) and 4), that $g_i \equiv X_i \pmod{X_r \mathfrak{m}_r}$ for $1 \leq i \leq r$ and $g_j \in X_r \mathfrak{m}_r$ for $j > r$. Applying elementary transformations to g again, we get that $g \equiv X(r) \pmod{X_r^2}$ by Proposition 5.

We now prove the second assertion. Suppose that $f = [f_1, \dots, f_r] \in E_1^t$ is $\mathfrak{m}_1$ -modular. Because $\mathfrak{m}_1/\mathfrak{m}_1^2$ is a free E_0 -module with the basis $X_1 + \mathfrak{m}_1^2$, we have

$$f_i \equiv \gamma_i X_1 \bmod X_1^2,$$

where $\gamma = [\gamma_1, \dots, \gamma_r] \in E_0^t$ is unimodular. By 3), we have $f^\beta \equiv X(1) \bmod X_1^2$ for some $\beta \in \mathbf{GL}(t, E_0)$. This proves the proposition.

PROPOSITION 7. *Suppose given an $\mathfrak{m}_r$ -modular vector $f \in E_r^t$, where $f \equiv X(r) \bmod X_r^2$. Then there is a unimodular vector $b \in E_r^t$, with $b \equiv e_r \bmod X_r$, such that $\langle b, f \rangle = X_r$.*

PROOF. We recall that

$$e_r = [0, \underbrace{\dots}_{r-1}, 0, 1, 0, \dots, 0].$$

Suppose that $f = [f_1, \dots, f_r]$, where $f_i = X_i + X_r^2 h_i$ for $1 \leq i \leq r$ and $f_j = X_r^2 h_j$ for $j > r$. There is a vector c such that $\langle f, c \rangle = X_r h_r$, whence $X_r = f_r - X_r \langle f, c \rangle$. We set $b = e_r - X_r c$. It follows from the preceding equation that $\langle f, b \rangle = X_r$ and therefore $(b) \ni X_r$ and $1 = b_r + X_r c_r$. This proves the proposition.

THEOREM 3. *$\mathbf{GL}(t, E_r)$ acts transitively on the set of $\mathfrak{m}_r$ -modular vectors in E_r^t for $t \geq r$.*

PROOF. Suppose that $r = 1$ and f is $\mathfrak{m}_1$ -modular. By Proposition 6, we may assume that $f \equiv X(1) \bmod X_1^2$. By Proposition 7, there is a unimodular vector $b \in E_1^t$ such that $\langle f, b \rangle = X_1$. By 3), $b = e_1^\alpha$ for some $\alpha \in \mathbf{GL}(t, E_1)$. Then, by (5), $(f^{\alpha*})_1 = X_1$. Moreover, we can reach $X(1)$ from $f^{\alpha*}$ by means of elementary transformations.

Suppose that the assertion is proved for all $i < r$ and that we are given an $\mathfrak{m}_r$ -modular vector $f \in E_r^t$. Reducing modulo X_r , we obtain an $\mathfrak{m}_{r-1}$ -modular vector $\bar{f} \in E_{r-1}^t$. By induction, $\bar{f}^\alpha = X(r-1)$ for some $\alpha \in \mathbf{GL}(t, E_{r-1})$. By Proposition 5 we may therefore assume that $f \equiv X(r) \bmod X_r$; that is, the conditions of Proposition 6 are satisfied. Consequently we may suppose that $f = X(r) \bmod X_r^2$. By Proposition 7, there is a unimodular vector $b \equiv e_r \bmod X_r$ such that $X_r = \langle b, f \rangle$. By 3) there is an $\alpha \in \mathbf{GL}(t, E_r)$ such that $b^\alpha = e_r$. By (5) we have $g_r = \langle e_r, g \rangle = X_r$ for $g = f^{\alpha*}$. Applying elementary transformations to g , we obtain a vector

$$c = [c_1, \dots, c_{r-1}, X_r, c_{r+1}, \dots, c_t], \quad c_i \in E_{r-1},$$

where $c' = [c_1, \dots, c_{r-1}, c_{r+1}, \dots, c_t]$ is $\mathfrak{m}_{r-1}$ -modular. By induction, $(c')^\beta = X(r-1)$ for some $\beta \in \mathbf{GL}(t-1, E_{r-1}) \subseteq \mathbf{GL}(t, E_r)$, because $t-1 \geq r-1$. This proves the theorem.

§3. Projective metabelian groups and Lie algebras

THEOREM 4. *If $n \geq 0$, then projective metabelian $A_n A$ -groups of finite rank are free.*

PROOF. By Theorem 2, we may assume that $n = q$ is either zero or prime. If P is a projective $A_q A$ -group of rank r , then a projective $\mathbf{Z}/(q)B_r$ -module S with an epimorphism $l: S \rightarrow \mathfrak{m}_r$ was constructed from P in §1.

LEMMA. *The conditions 1)–4) of §2 are satisfied for $E_r = \mathbf{Z}/(q)B_r$.*

PROOF. We have a chain $E_0 = \mathbb{Z}/(q) \subset E_1 \subset \dots \subset E_r \subset \dots$, where $1 \in E_0$ and E_r is a retract of E_{r+1} with kernel $(x_{r+1} - 1)$. By results of Suslin [7], each $\mathrm{GL}(t, E_r)$ acts transitively on the unimodular vectors of E_r^t for all $t \geq 1$.

It remains to verify 4). Suppose that $\mathfrak{m}_r = (X_1 - 1, \dots, X_r - 1)$ and we have

$$\sum_{i=1}^r a_i (X_i - 1) + \sum_{r \geq i \geq j \geq 1} b_{ij} (X_i - 1) (X_j - 1) = 0 \quad (7)$$

for $a_i \in \mathbb{Z}/(q)$ and $b_{ij} \in E_r$. Setting $X_j = 1$ for all $j > 1$ in (7), we obtain

$$a_1 (X_1 - 1) + b_{11} (X_1 - 1)^2 = 0.$$

But E_r is an integral domain, and therefore $a_i \in \mathbb{Z}/(q) \wedge \mathfrak{m}_r = 0$; that is, $a_1 = 0$. Similarly, $a_2 = \dots = a_r = 0$ in (7). This proves the lemma.

We now turn to the proof of the theorem. We remark that the module S constructed in §1 is of rank r ([9], Chapter II, §5). In fact, because $x_i^\pi \equiv x_i \pmod{H'}$ and $y_j^\pi \in H'$, we have $u_i^\rho \equiv u_i \pmod{\mathfrak{m}_r M}$, $v_j^\rho \in \mathfrak{m}_r M$ and S is therefore generated modulo $\mathfrak{m}_r S$ by the elements $u_i + \mathfrak{m}_r S$, from which our assertion follows. Thus, by a result of Suslin and Theorem 3, S has a free basis $t_1, \dots, t_r$ such that $l(t_i) = X_i - 1$. By Proposition 2, H contains the elements

$$a_i = \begin{pmatrix} X_i & 0 \\ t_i & 1 \end{pmatrix}, \quad i = 1, \dots, r.$$

But because $t_i^\rho = t_i$, we have $a_i^\pi = a_i \in P$ by Proposition 3. It remains to show that $a_1, \dots, a_r$ generate P . Suppose that

$$\begin{pmatrix} b & 0 \\ c & 1 \end{pmatrix} \in P, \quad b = X_1^{n_1} \dots X_r^{n_r}, \quad c = c^\rho = \sum q_i t_i.$$

Multiplying by $\Pi_1^r a_i^{-n_i}$, we may assume that $b = 1$. Then $l(c) = \sum q_i (X_i - 1) = 0$ and, by Proposition 4, c belongs to the submodule generated by $t_i(X_j - 1) - t_j(X_i - 1)$; that is, by (2'),

$$\begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} = \prod [a_i, a_j]^{B_{ij}}.$$

Thus $a_1, \dots, a_r$ generate P . Because the rank of P is r , they generate P freely. This proves the theorem.

We now consider projective metabelian Lie algebras.

THEOREM 5. Suppose that $k = k_0[Y_1^{\pm 1}, \dots, Y_n^{\pm 1}, Z_1, \dots, Z_m]$, where k_0 is a principal ideal ring. Then projective metabelian Lie k -algebras of finite rank are free.

PROOF. We remark that all the results of §1 of [4] also hold for an arbitrary commutative ring k , since a projective metabelian Lie k -algebra is a free k -module, a basis of which is formed by the proper left-normed monomials. Moreover, the assertion (B) $\Rightarrow$ (A) in Theorem 1 of [4], §2, was also proved for an arbitrary commutative k . It is therefore enough for us to show that any projective $k[X_1, \dots, X_r]$ -module S , $r(S) = r$, with an

epimorphism $l: S \rightarrow m_r = (X_1, \dots, X_r)$ is free and has a basis $t_1, \dots, t_r$ such that $l(t_i) = X_i$. But this follows from Theorem 3 by the results of Suslin [7], if we put $k[X_1, \dots, X_r]$ in the place of E_r . This proves the theorem.

COROLLARY. *Suppose that k is a field, with $\text{char } k = p > 0$. Then projective restricted metabelian Lie k -algebras of finite rank are free.*

The proof follows from the results of [6] by Theorem 5.

Received 1/SEPT/1976

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Translated by A. J. McISAAC